

The Migration Hump Paradox

Second Paper

Raphaël Henry & Julian Ansarinasab

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This Second Paper uses the same data file as the First Paper, migration_data.gdt, and exactly the same dependent variable. Nothing is redefined here, so all the results below are directly comparable with those of the First Paper.

1. Selection of the Optimal Model

To determine which variables should be retained in our final model, we will employ two distinct methodologies discussed in class to identify the most robust specifications for our project.

The first method is the "All Possible Regressions" approach. While this technique is computationally intensive, given our set of four explanatory variables, it offers the significant advantage of evaluating every possible combination of predictors within the set.

Our selection process follows a two-step refinement:

- **Significance Filter:** We first estimate all potential models and retain only those in which every included variable is statistically significant.
- **Criterion Optimization:** Among the remaining candidates, we select the single model that minimizes the Schwarz Criterion (SC).

We have prioritized the Schwarz Criterion over the Akaike Information Criterion (AIC) because it is mathematically more "consistent". As the sample size (n) increases, the SC is more likely to converge on the true, most accurate model. Furthermore, the SC acts as a stricter judge of model quality; because the penalty for adding extra variables becomes increasingly heavy as the sample grows, it effectively guards against over-specification. The penalty term is $[k \times \ln(n)]/n$; it represents the cost of adding variables.

The first step is to create the 15 different models and determine among them which have all their explanatory variables significant.

gretl : sortie de code

```

? ols Y const X1
? matrix c = {1, $aic, $bic}
? ols Y const X2
? matrix c = c | {2, $aic, $bic}
( ... 15 modèles ... )
? colnames(c, "model AIC SC")
? print c
c (15 x 3)
      model      AIC      SC
      1      858,6054  864,1803
      2      844,9547  850,5297
      3      855,6452  861,2202
      4      856,6110  862,1860
      5      842,1334  850,4959
      6      857,4231  865,7856
      7      858,4711  866,8336
      8      846,1040  854,4665
      9      846,8872  855,2497
     10      855,8527  864,2152
     11      843,0023  854,1523
     12      844,0923  855,2423
     13      857,4436  868,5935
     14      848,0250  859,1750
     15      844,9514  858,8888

```

The p-values of the slopes of the 15 models are reported below. A model passes the significance filter only if every one of its slopes is significant (shown in bold).

Model	Variables	p-values of the slopes	Retained
1	X1	0.8526	-
2	X2	0.0003	yes
3	X3	0.0868	yes
4	X4	0.1586	-
5	X1 X2	0.0305 / 0.0000	yes
6	X1 X3	0.6424 / 0.0788	-
7	X1 X4	0.7125 / 0.1500	-
8	X2 X3	0.0008 / 0.3635	-
9	X2 X4	0.0007 / 0.7980	-
10	X3 X4	0.1017 / 0.1871	-
11	X1 X2 X3	0.0267 / 0.0001 / 0.2968	-
12	X1 X2 X4	0.0317 / 0.0001 / 0.8423	-
13	X1 X3 X4	0.5303 / 0.0878 / 0.1675	-
14	X2 X3 X4	0.0021 / 0.3623 / 0.7827	-
15	X1 X2 X3 X4	0.0278 / 0.0002 / 0.2968 / 0.8255	-

p-values of the slopes of the 15 estimated models

⇒ Model 1 is not significant, unlike model 2, which we retain for now.

⇒ Models 2, 3 and 5 pass the significance filter (model 3 only at the 10% level); all the models after model 5 are rejected because at least one of their variables is not significant.

Among all the relevant models, we choose the one with the smallest SC. On Gretl we display the two criteria (AIC and BIC/SC) to check whether they would give the same result at the end.

gretl : sortie de code					
Modèles retenus après le filtre de significativité					
model	variables	AIC	SC		
2	X2	844,95	850,53		
3	X3	855,65	861,22		
5	X1 X2	842,13	850,50	<-- SC minimum	

The Optimal Model is the fifth model, composed of only the following explanatory variables: Net Official Development Assistance and the GDP per capita in PPA. It minimises both the AIC (842.13) and the SC (850.50), even though the gap with model 2 on the Schwarz criterion is very small (850.50 against 850.53), the two criteria pointing to the same specification.

Let's conduct a second test to see if Model 5 is still the Optimal Model. The second methodology is Backward Elimination. We begin by estimating the complete model, incorporating all four explanatory variables.

The elimination process is guided by the following logic:

- **Significance Check:** If any variables are found to be statistically non-significant, we proceed with a removal step.
- **Selection for Removal:** In cases where multiple variables are non-significant (as observed in our initial Gretl model) we identify the candidate for rejection by looking for the highest p-value or the lowest Student's t-statistic.
- **Iterative Refinement:** We re-estimate the model without the rejected variable and repeat this cycle until every remaining predictor in the model is statistically significant.

gretl : sortie de code											
<pre> # Method 2 - Backward Elimination # Start with the full model ? ols Y const X1 X2 X3 X4 const 10,5656 1,92172 5,498 2,35e-07 *** X3 -0,0568885 0,0542761 -1,048 0,2968 X1 -0,259355 0,116378 -2,229 0,0278 ** X4 -0,194182 0,878703 -0,221 0,8255 X2 -0,000395900 0,000103042 -3,842 0,0002 *** # we eliminate the variable with the higher p-value, so X4 ? ols Y const X1 X2 X3 const 10,7355 1,75383 6,121 1,30e-08 *** X1 -0,260048 0,115858 -2,245 0,0267 ** X2 -0,000402878 9,76814e-05 -4,124 0,0001 *** X3 -0,0566425 0,0540418 -1,048 0,2968 # we eliminate the only variable that is not significant, so X3 ? ols Y const X1 X2 const 9,94684 1,58486 6,276 6,08e-09 *** X1 -0,253472 0,115736 -2,190 0,0305 ** X2 -0,000422887 9,58382e-05 -4,413 2,29e-05 *** # All the variables are now significant, we stop here # => we get the same model as with the All Possible Regressions </pre>											

Notably, both the "All Possible Regressions" and "Backward Elimination" methods converged on the same optimal model. This consistency allows us to confidently conclude that Official Development Assistance (ODA) and GDP per capita (PPP) are the primary drivers of our migration indicator. While our other two variables (Unemployment and Political Stability) showed some influence during the initial descriptive analysis, they lack sufficient statistical power in the multiple regression and can therefore

be omitted from the final specification.

gretl : sortie de code

```
? ols Y const X1 X2          # model 5 (optimal)
Modèle 5: MCO, utilisant les observations 1-120
Variable dépendante: Y
```

	coefficient	éc. type	t de Student	p. critique	
const	9,94684	1,58486	6,276	6,08e-09	***
X1	-0,253472	0,115736	-2,190	0,0305	**
X2	-0,000422887	9,58382e-05	-4,413	2,29e-05	***
Moyenne var. dép.	4,013619		Éc. type var. dép.	8,552710	
Somme carrés résidus	7460,616		Éc. type régression	7,985359	
R2	0,142922		R2 ajusté	0,128271	
F(2, 117)	9,755179		P. critique (F)	0,000121	
Log de vraisemblance	-418,0667		Critère d'Akaike	842,1334	
Critère de Schwarz	850,4959		Hannan-Quinn	845,5297	

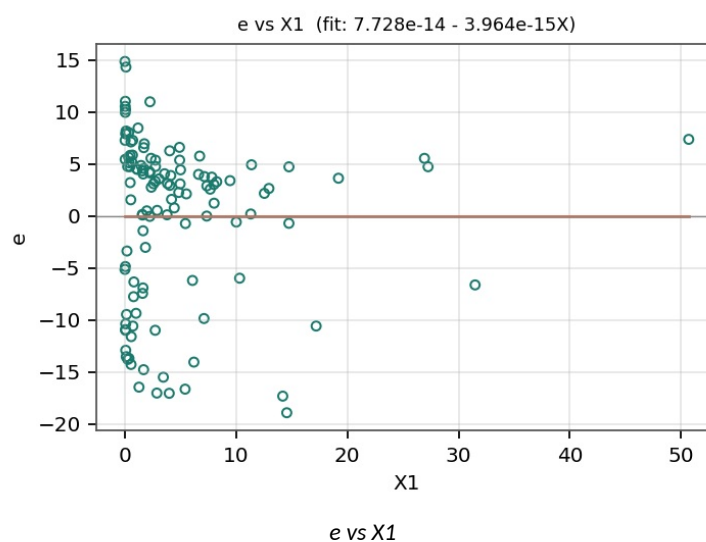
2. Tests for Detecting Error Autocorrelation

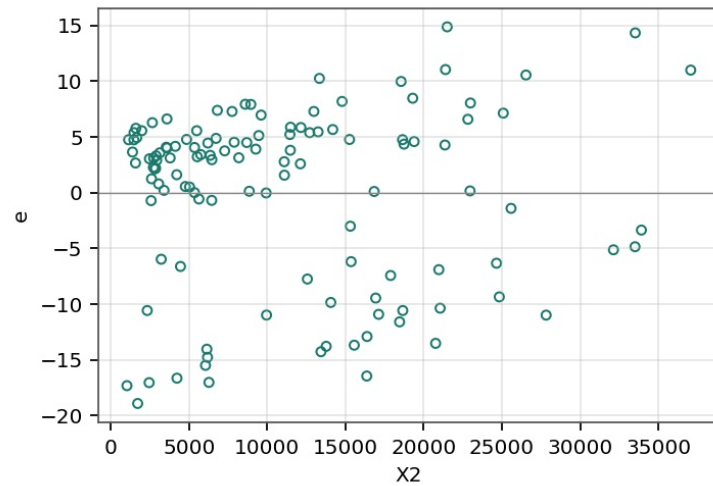
To detect the autocorrelation of errors, we will apply three distinct diagnostic methods: visual examination of estimation residuals, the Durbin-Watson (DW) test, and the Breusch-Godfrey test.

From this point forward, our analysis will focus exclusively on our Optimal Model to verify whether it adheres to the H5 OLS assumption. This assumption states that residuals must be uncorrelated; a violation here implies that the model's reliability is compromised, necessitating corrective measures in the final stage of this project.

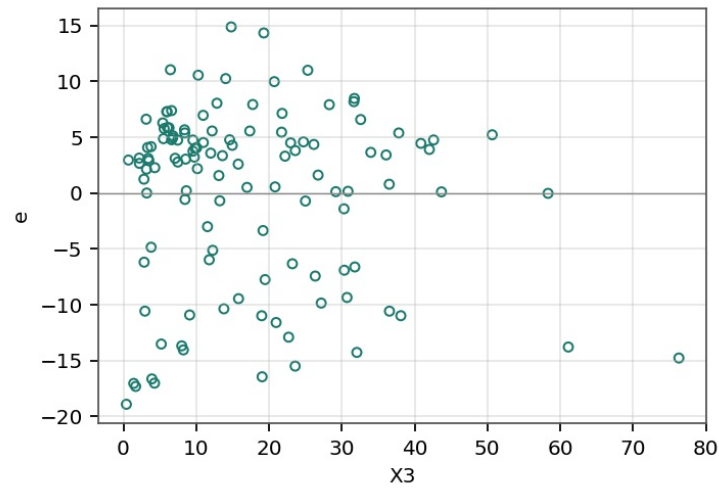
Visual Examination of Residuals:

For cross-sectional data, the detection of autocorrelation requires a specific approach. Unlike time-series data where residuals are plotted against time, we must plot our residuals against each of the explanatory variables. The goal of this visual inspection is to identify which specific variable might be causing the error terms to be correlated. Once a suspect variable is identified through these plots, it will serve as the basis for performing the formal Durbin-Watson test.

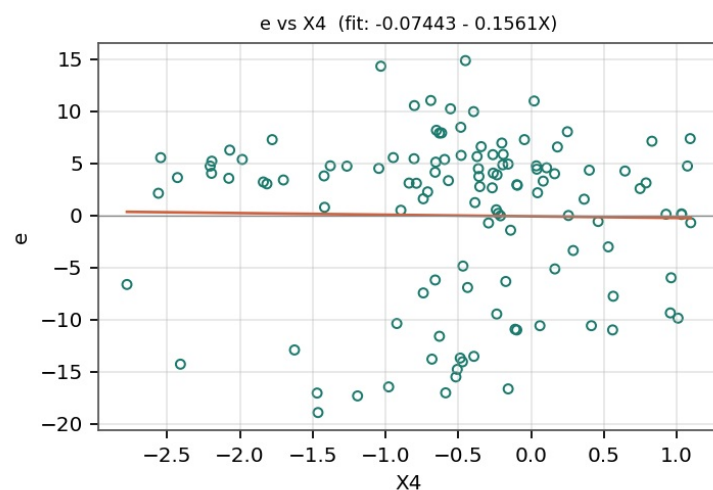




e vs X2



e vs X3



e vs X4

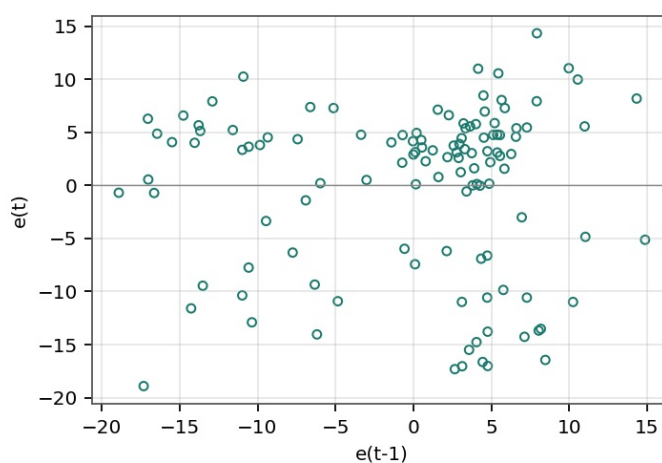
- ⇒ A slight orientation of the cloud appears when ordering by X1.
- ⇒ We don't observe any correlation between the residuals when ordering by X2 and X3, meaning no autocorrelation.

⇒ We can observe a small orientation when ordering by X4.

The most flagrant pattern is obtained with X1, which leads us to assume that X1 is the most likely source of an autocorrelation of errors. By sorting the dataset in ascending order of X1, we can effectively treat our cross-sectional data like a time series and apply standard diagnostic tests. This includes using the Durbin-Watson test to check for first-order autocorrelation and the Breusch-Godfrey test for higher-order patterns. Between the two, the Breusch-Godfrey test is the more reliable choice because it eliminates the "inconclusive zone" often found in Durbin-Watson results, allowing for a much clearer interpretation of our model's validity.

Important: all the tests of this section are computed after the instruction `dataset sortby X1`, so that the "time" ordering used by the DW and BG statistics is indeed the ordering by X1.

Scatter plot of $e(t)$ as a function of $e(t-1)$, after sorting by X1:



gretl : sortie de code

```
? dataset sortby X1
? ols Y const X1 X2
? series e = $uhat
? series e_1 = e(-1)
? series diff2 = (e - e_1)^2
? scalar sumee1 = sum(diff2)
? scalar sume = sum(e^2)
? scalar DW = sumee1 / sume
--- DURBIN-WATSON ---
DW = 1,95405
# k = 2 explanatory variables, n = 120 : d1 = 1,63 ; d2 = 1,72
# 4 - d2 = 2,28 ; 4 - d1 = 2,37
```

⇒ $DW = 1.954$ lies between $d2 = 1.72$ and $4 - d2 = 2.28$: according to the Durbin-Watson decision rule we accept H_0 , there is no autocorrelation in our model.

Let's conduct a Breusch-Godfrey test with an autocorrelation order equal to 1.

Test Hypothesis:

- H_0 : no autocorrelation between residuals (error terms are independent)
- H_1 : there is an autocorrelation

Test Statistics: $LM = n'R^2$ follows a chi-square(p), with p: degree of autocorrelation and $n' = n - p$

Rejection Region: $LM > \chi^2(p)$

gretl : sortie de code

```
? ols e const X1 X2 e_1
Modèle 6: MCO, utilisant les observations 2-120 (n = 119)
Variable dépendante: e
```

	coefficient	éc. type	t de Student	p. critique
const	0,0613059	1,58018	0,0388	0,9691
X1	0,00253057	0,114963	0,0220	0,9825
X2	-1,77350e-05	9,63170e-05	-0,1841	0,8542
e_1	0,00717121	0,0930424	0,0771	0,9387
R2		0,000434	R2 ajusté	-0,025687

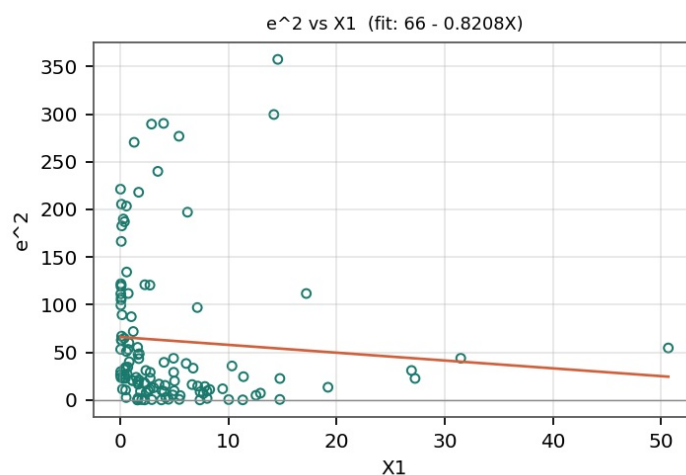
```
? scalar n1 = $nobs - 1
? scalar LM = n1 * $rsq
? scalar crit = critical(X, 1, 0.05)
--- BREUSCH-GODFREY ---
LM = 0,0516 | Chi2(1) at 5% = 3,8415
```

⇒ $LM = 0.0516 < 3.841$: we cannot reject H_0 . Therefore, we proved that there is no autocorrelation in our Optimal Model, which confirms the conclusion of the Durbin-Watson test.

3. Tests for Detecting Error Heteroscedasticity

The fact that variances of disturbances are no longer constant is a common problem in cross-section data, due to the fact that the observations represent averages calculated on samples of different sizes; that the same value of the explanatory variable is repeated for different values of the endogenous variable; and that errors may be related to the values taken by an explanatory variable.

We will start by doing the scatterplot of squared residuals against the variable suspected to cause the problem, X1:



Contrary to what a rapid visual inspection could suggest, the adjustment line of e^2 on $X1$ has a slightly negative slope (-0.8208) which is not statistically significant ($t = -0.804$). The dispersion of the squared residuals therefore does not increase systematically with the level of aid. This first visual diagnostic must nevertheless be confirmed by formal tests: we will perform the Goldfeld-Quandt and the White tests. The Goldfeld-Quandt test is especially relevant here, as it is designed for cases where a specific variable is the suspected cause of heteroskedasticity within a sufficiently large sample.

Goldfeld-Quandt test

The observations are sorted in ascending order of X_1 ; we remove the central quarter of the sample ($c = 30$ observations) and estimate the Optimal Model on the two remaining sub-samples of 45 observations each.

gretl : sortie de code

```
? scalar n_GQ    = $nobs
? scalar c_obs    = floor(n_GQ/4)
? scalar N_group  = floor((n_GQ - c_obs)/2)
? smpl 1 N_group
Plage d'observations restreinte, 45 observations conservées
? ols Y const X1 X2
? scalar ssr1 = $ess    # 3584,3630
? scalar df1 = $df      # 42
? smpl full
? smpl S2_start n_GQ
Plage d'observations restreinte, 45 observations conservées
? ols Y const X1 X2
? scalar ssr2 = $ess    # 2230,6736
? scalar df2 = $df      # 42
? smpl full
? scalar GQ      = (ssr2/df2)/(ssr1/df1)
? scalar GQ_crit = critical(F, df2, df1, 0.05)
--- GOLDFELD-QUANDT ---
GQ = 0,6223 | F* = 1,6710 | p-value = 0,9358
```

⇒ $GQ = 0.6223 < 1.6710$: we do not reject H_0 , the variance of the errors does not increase with X_1 .

Because the ratio is below 1, we also tested the reverse alternative (a variance *decreasing* with the level of aid): the inverted ratio is 1.6069 with a p-value of 0.0642, which is still above the 5% threshold. The Goldfeld-Quandt test therefore does not allow us to conclude that heteroskedasticity is present, in either direction.

White test

We will now proceed with a White Test, which has a dual purpose: it detects and informs us about the form of the heteroskedasticity. The test is based on the relationship between the squared residual from the estimation of Y on X and one or more variables x_k suspected to be the source of the problem.

gretl : sortie de code

```

? series e2 = e*e
? series X1_2 = X1 * X1
? series X2_2 = X2 * X2
? series X1X2 = X1 * X2
? ols e2 const X1 X2 X1_2 X2_2 X1X2
Modèle 7: MCO, utilisant les observations 1-120
Variable dépendante: e2

```

	coefficient	éc. type	t de Student	p. critique	
const	40,2459	30,9837	1,299	0,1966	
X1	1,32856	3,33672	0,398	0,6913	
X2	0,00370057	0,00372906	0,992	0,3231	
X1_2	0,0601235	0,0757960	0,793	0,4293	
X2_2	-6,00880e-08	1,01930e-07	-0,590	0,5567	
X1X2	-0,000733519	0,000367004	-1,999	0,0480	**
R2	0,052847	R2 ajusté	0,011272		

```

? scalar LM_white = $nobs * $rsq
? scalar df_white = $ncoeff - 1
? scalar crit_white = critical(X, df_white, 0.05)
? scalar pval_white = pvalue(X, df_white, LM_white)
--- WHITE TEST ---
R2 aux = 0,052847 | LM = 6,3417 | Chi2(5) = 11,0705 | p-value = 0,2744

```

N.B: The degrees of freedom (df) for the White test are determined by the number of independent variables in the auxiliary regression, excluding the constant term. In our model, the auxiliary equation includes the two original regressors (X1 and X2), their squared terms ($X1^2$ and $X2^2$), and their cross-product interaction (X1X2), totaling five distinct restrictions. Therefore, the test statistic follows a Chi-square distribution with 5 degrees of freedom and at a 5% significance level the critical value is 11.0705.

Statistical Evidence

At a 5% significance level, with a sample size of $n = 120$ and an auxiliary R^2 of only 0.0528, the calculated Lagrange Multiplier (LM) statistic reaches 6.3417. This value remains well below the Chi-square critical value of 11.0705 (p -value = 0.2744). The independent variables, their squares and their interactions explain barely 5% of the variance of the squared residuals: we **cannot reject** the null hypothesis (H_0) of homoscedasticity. Only the cross-product X1X2 is individually significant ($p = 0.0480$), which is not sufficient to conclude in favour of a heteroskedastic structure since the joint test is largely non-significant. The built-in test of Gretl (`modtest --white`) gives exactly the same conclusion.

Interpretation

- **Assumption H4 satisfied:** neither the Goldfeld-Quandt test nor the White test detects a non-constant variance of the disturbances in the Optimal Model.
- **Valid inference:** the OLS standard errors, and therefore the t-statistics, the p-values and the confidence intervals reported in section 1 and in the First Paper, are valid.
- **BLUE estimators:** combined with the absence of autocorrelation established in section 2, this means that the OLS estimator of the Optimal Model is indeed the Best Linear Unbiased Estimator.

4. Correction for error heteroskedasticity

To correct heteroskedasticity we would apply the Generalized Least Squares Method (GLS), which is called Weighted Least Squares in the case of errors that are heteroskedastic. If more than one variable

is significant in the White auxiliary equation, we select the most significant variable (highest t-ratio) and divide every other variable (dependent and explanatory variables and the constant term) by the square root of this variable in order to restore homoskedasticity in the model.

Applied to our specification, the full auxiliary model would be:

$$Y = a_0 + a_1 X_1 + a_2 X_2 + a_3 (X_1)^2 + a_4 (X_2)^2 + a_5 X_1 X_2$$

and, if $(X_1)^2$ had been identified as the source of the non-constant variance, each variable would have been multiplied by $1/\sqrt{(X_1)^2} = 1/X_1$, giving:

$$Y/X_1 = a_0/X_1 + a_1 + a_2 X_2/X_1 + a_3 (X_1)^2/X_1 + a_4 (X_2)^2/X_1 + a_5 X_1 X_2/X_1$$

In our case this correction is not required, and it would even be harmful. First, the joint White test does not reject homoskedasticity ($LM = 6.34 < 11.07$), so there is no non-constant variance to correct. Second, the weighting variable $1/X_1$ is not admissible here: the variable X_1 (net ODA in % of GNI) takes values extremely close to zero, and even slightly negative for one country (minimum = -0.0015), so that the weights $1/X_1$ explode and change sign. A GLS transformation by $1/X_1$ would mechanically produce a very high R^2 and apparently "clean" residuals while in fact destroying the economic content of the regression.

Robustness check: White's heteroskedasticity-robust standard errors

As a precaution, and because the cross-product $X_1 X_2$ was individually significant in the auxiliary equation, we re-estimated the Optimal Model with White's heteroskedasticity-robust standard errors. The coefficients are unchanged by construction; only the standard errors are affected.

gretl : sortie de code

```
? ols Y const X1 X2 --robust
```

Modèle 8: MCO, utilisant les observations 1-120

Variable dépendante: Y

Éc. type robustes à l'hétéroscédasticité, variante HC1

	coefficient	éc. type	t de Student	p. critique	
const	9,94684	1,49014	6,675	8,40e-10	***
X1	-0,253472	0,106089	-2,389	0,0185	**
X2	-0,000422887	9,89000e-05	-4,274	3,90e-05	***
Moyenne var. dép.		4,013619	Éc. type var. dép.	8,552710	
Somme carrés résidus		7460,616	Éc. type régression	7,985359	
R2		0,142922	R2 ajusté	0,128271	

⇒ Both variables remain significant, and X_1 is even slightly more significant than with the usual OLS standard errors ($p = 0.0185$ instead of 0.0305). The conclusions of the First Paper are therefore not statistical artifacts of a badly specified variance: they are robust to the treatment of the error variance.

5. Conclusion

This research aimed to investigate the Migration Hump Paradox by analysing the relationship between Official Development Assistance (ODA), GDP per capita and the migration balance across 120 countries. Through a rigorous econometric selection process involving both All Possible Regressions and Backward Elimination, we identified ODA (X_1) and GDP (X_2) as the primary significant drivers of our migration indicator, the two methods converging unanimously on the same specification (Model 5, SC = 850.50).

We then verified that this model satisfies the classical assumptions of the linear model. The Durbin-Watson statistic ($DW = 1.954$, between $d2 = 1.72$ and $4 - d2 = 2.28$) and the Breusch-Godfrey test ($LM = 0.0516$, far below the critical value 3.841) both conclude to the absence of autocorrelation of the errors. The Goldfeld-Quandt test ($GQ = 0.6223$, critical value 1.6710) and the White test ($LM = 6.3417$, critical value 11.0705) do not reject homoskedasticity either. The OLS estimator of the Optimal Model is therefore BLUE and no GLS correction is required; the robust standard errors confirm the significance of both variables.

Ultimately, the two papers deliver a consistent message: once the whole project is estimated on a single data file and a single definition of the dependent variable, aid and wealth are the only two significant determinants of our migration indicator, and the effect of aid is stable across the poorest and the richest countries of the sample (Chow test of the First Paper, and non-significant interaction term with the D_Rich dummy, $p = 0.9756$).

These results must nevertheless be read with two precautions in mind. The dependent variable is a sign-preserving logarithmic transformation of the migration ratio, so the coefficients measure an effect on that transformed indicator and not an elasticity of the number of migrants; and the use of a net migration balance can mask the emigration of a country that simultaneously receives regional flows. Our results therefore suggest that, for this specific sample, the link between international aid and migration flows is real but fragile, and that any policy conclusion drawn from a cross-sectional model of this kind should remain prudent.